IMA FAST TRACK
6 MONTHS A/L
ENGLISH MEDIUM

Economics

Market Demand Supply and Equilibrium

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GCE ADVANCED LEVEL ENGLISH MEDIUM

❒ Demand ❒

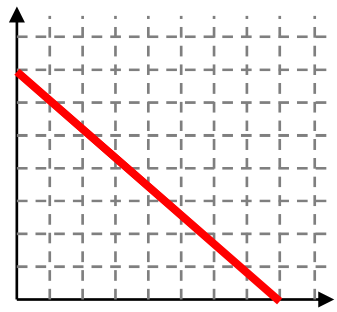
In a certain period of time, when other factors remain constant, various quantities ready to buy at various prices of a good is called as demand.

⏩ ⏩ The deference between demand and wants

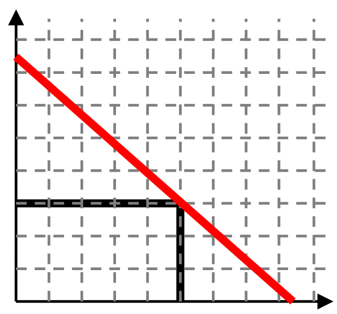
- The various forms of fulfilling needs are called wants.
- Because of the central economic problem of scarcity, cannot be achieved every wants.
- Accordingly, demand can be identified as the wants that decide to fulfill among unlimited wants.

⏩ ⏩ The deference Demand and Quantity demand

- In a certain period of time, when other factors remain constant, various quantities ready to buy at various prices of a good is called as demand.
  - Demand vcan be represented by a demand curve.



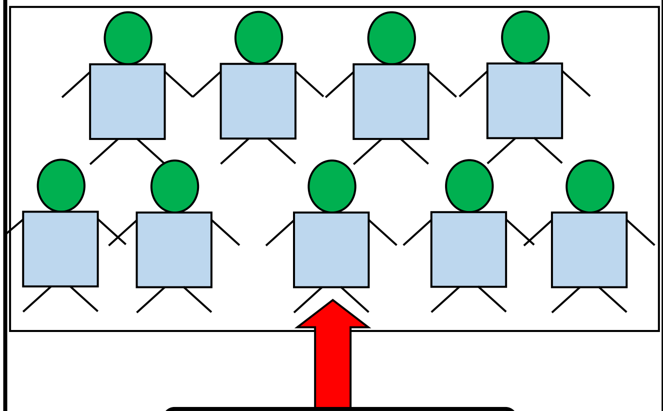
- The quantity demand can be defined as the quatity purchased at a certain price.
  - Quantity demand can be represented by a point on the demand curve.



DEMAND

Individual demand

At a certain period of time, the various quantities ready to buy by a person at various prices of good is individual demand.



only one person

Determinants of Individual demand

- Price of the concerned goods(P)
- Price of the related goods(Pn)
- Consumer Income(Y)
- Consumers' Taste (T)
- Future expectations (Ex)
- Other factors (O)

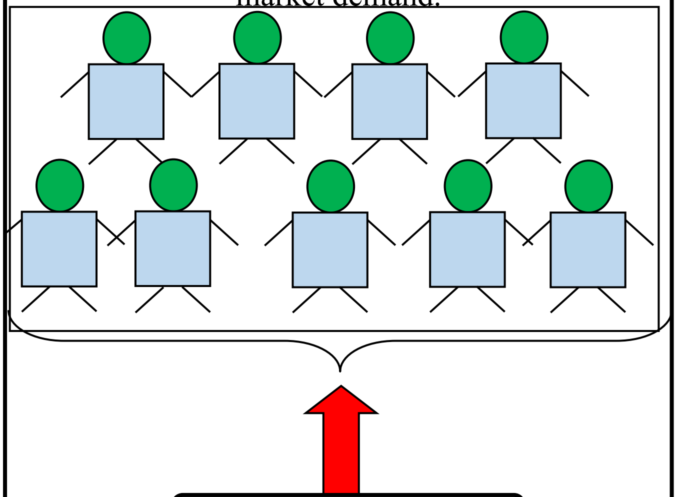
Individual demand function.

The relationship between individual demand and determinants of demand as an equation it is named as individual demand function.

$$Qd = f(P, Pn, Y, T, Ex, O)$$

Market demand

At a certain period of time,the various quantities ready to buy by sum of all individuals at various prices of good is called as market demand.



sum of every individuals

Determinants of Marketdemand

- Price of the concerned goods(P)
- Price of the related goods(Pn)
- Consumer Income(Y)
- Consumers' Taste (T)
- Future expectations (Ex)
- Number of buyers and the composition (N)
- Other factors (O)

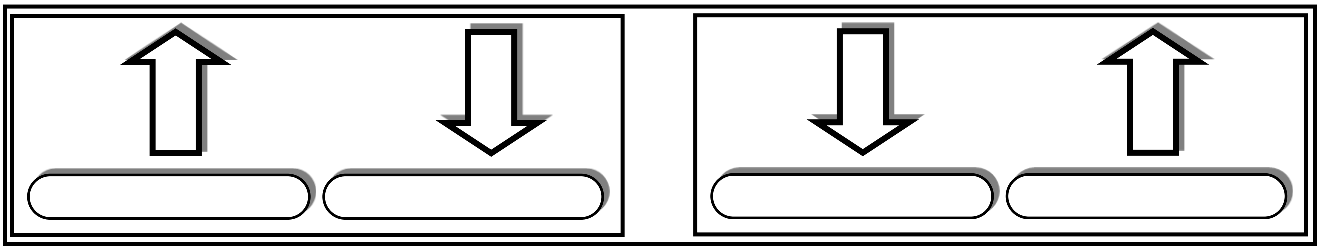
Market demand function.

The relationship between market demand and the determinants of market demand as an equation it is named as Market demand function.

$$Qd = f(P, Pn, Y, T, Ex, N, O)$$

Analysis of the changes of demand according to the changes of any determinants of demand is called Theory of demand.

In a certain period of time, when other factors affecting demand remains constant the inverse relationship between price and the quantity demand of goods, is called the law of demand.



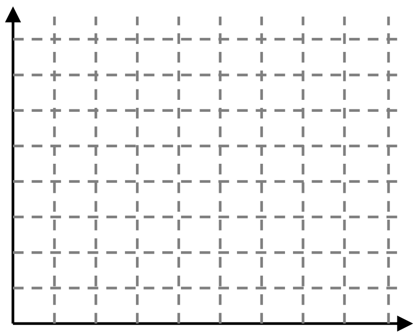
Assumptions

- Considers price and quantity demanded at a certain time period
- Other factors affecting demand remain constant except the price
- Considers demand of a normal good
- Considers behavior of rational buyers

The alternative methods which represent The Law of Demand are as follows.

- Demand schedule
- Demand curve
- Demand equation

Price	Quantity D
0	300
10	200
20	100
30	0

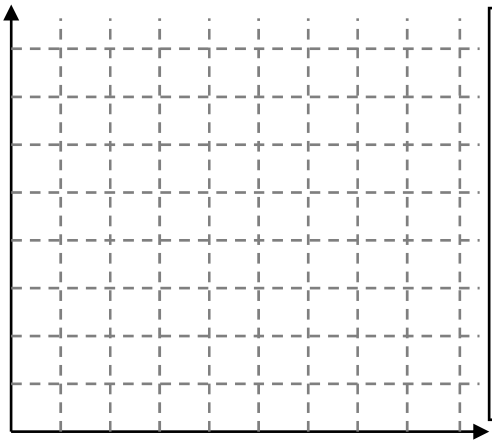


$$QD= a-bP$$

❍ Make the demand curve and estimate the demand equation.

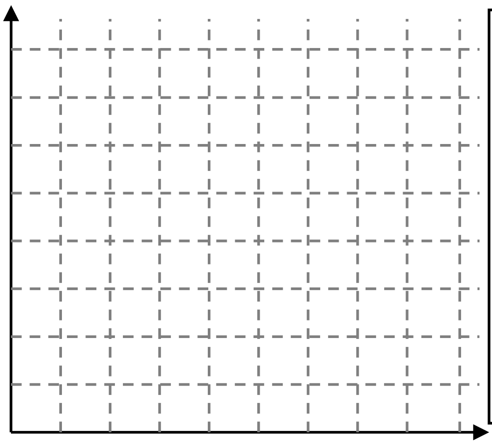
1)

මිල(P)	ප්‍රමාණය(Qd)
0	300
10	200
20	100
30	0



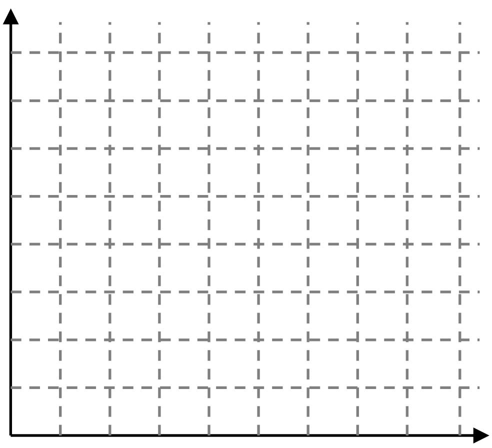
2)

මිල(P)	ප්‍රමාණය(Qd)
10	200
20	160
30	120
40	80



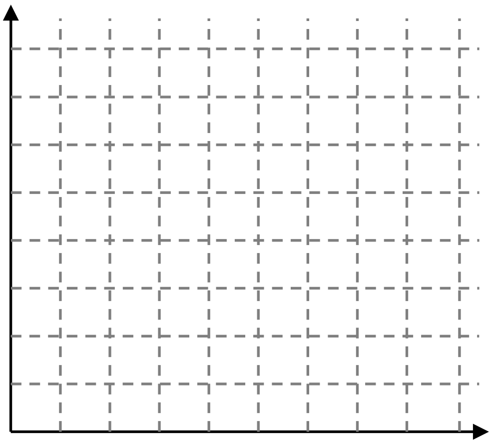
3)

මිල(P)	ප්‍රමාණය(Qd)
2	16
4	12
6	8
8	4



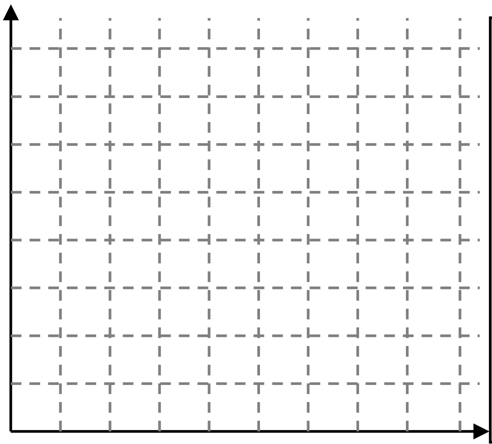
4)

මිල(P)	ප්‍රමාණය(Qd)
5	75
10	50
15	25
20	0



5)

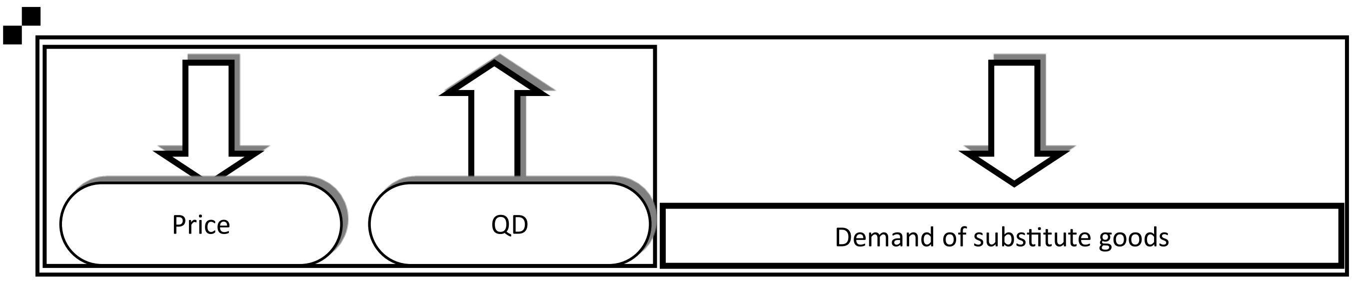
මිල(P)	ප්‍රමාණය(Qd)
10	80
15	60
20	40
25	20

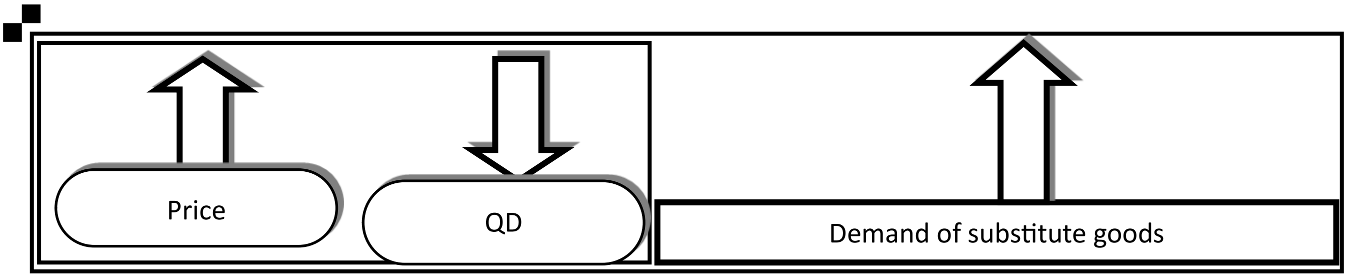


Reasons for the inverse relationship between price and quantity demand

❒ Substitution effect of a price change

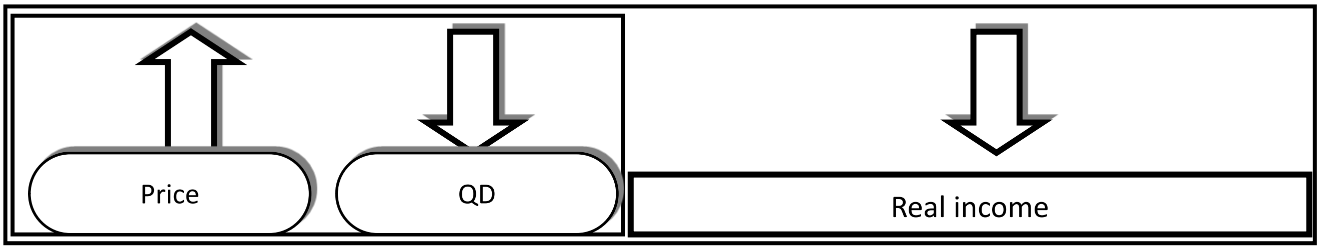
When other factors remain constant including the price of substitute goods, due to increase or decrease of the price of concerned goods, change of quantity demand of the considering good as a result of increase or decrease in relative price of the good is known as substitute effect.

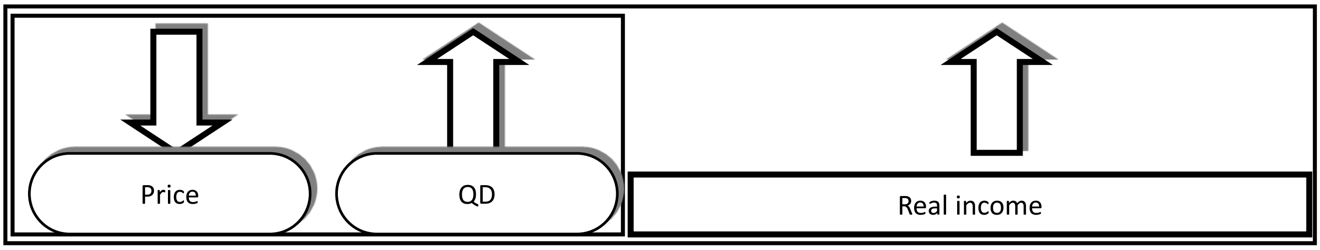




❒ Income effect of a price change

When other factors remain constant including the nominal income of consumers, changes of the quantity demanded according to the changes of real income is called income effect.





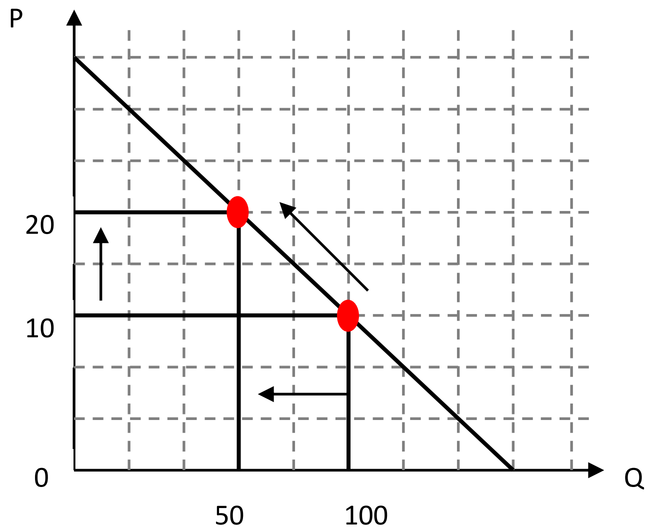
⏩ Changing quantity demand ⏩

When the other factors constant remain, except the price of concerned goods, changes of the quantity demanded in response to the increase or decrease of the price of the considering good is called changes of quantity demanded.

⏩ **When other factors are constant, <u>if the price of the concerned good is increased</u> the quantity demanded is decreased.**

- The point on the demand curve move upward along the demand curve.

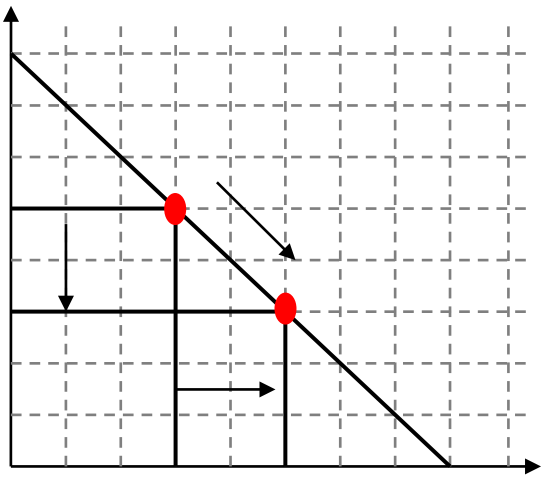
This situation is defined as a <u>contract demand</u>.



⏩ **When other factors are constant, if the price of the concerned good <u>decrease</u> the quantity demanded increases.**

- The point on the demand curve moves downward along the demand curve.

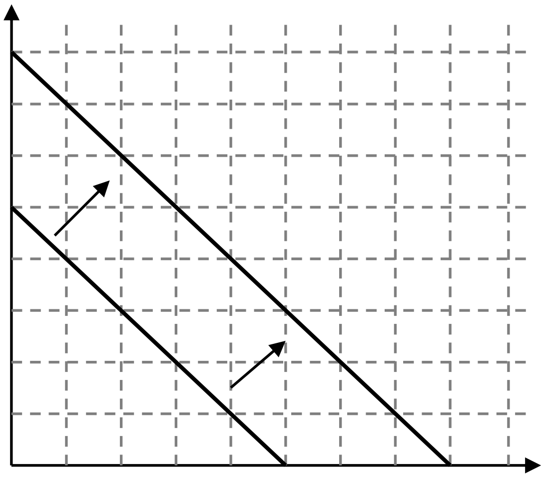
This situation is defined as <u>expansion demand</u>.



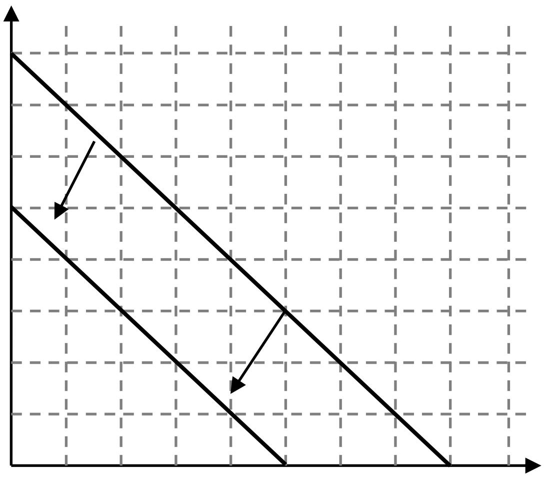
⏩ Changing quantity demand ⏩

⏩ **When the price of the concerned goods remains constant, change of demand according to the change of other factors is known as change in demand.**

- Increase of the demand response to the changes of other factors of demand while the price of the concerned goods remains constant is called increase in demand and demand curve shifts to the right.



- Decrease of the demand response to the changes of other factors of demand while the price of the concerned goods remains constant is called decrease in demand and demand curve shift to the left.



Factors for shift of the demand curve to the Right.	Factors for shift of the demand curve to the Left.
• Increase of the price of substitute goods	• Decrease of the price of substitute goods
• Decrease of the price of complementary goods	• Increase of the price of complementary goods
• Increase of the consumer income	• Decrease of the consumer income

• Increase of the consumer taste	• Decrease of the consumer taste
• Expect that the price will increase in the future	• Expect that the price will decrease in the future

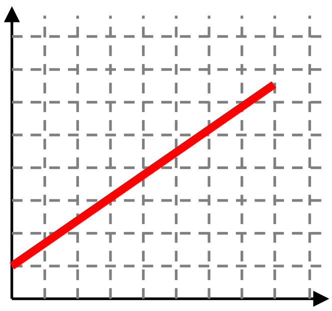
❒ SUPPLY ❒

At a certain period of time the various quantities that the producers are willing and able to supply, at alternative prices is called supply.

⏩ ⏩ The deference Supply and Quantity Supply

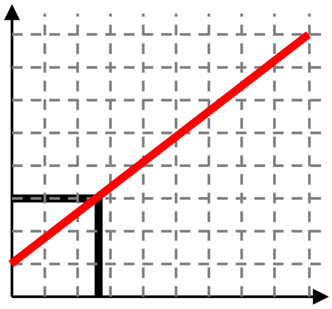
▪In a certain period of time, when other factors remain constant, various quantities ready to supply at various prices of a good is called as demand.

- Supply can be represented by a supply curve.



▪The quantity supply can be defined as the quatity supply at a certain price.

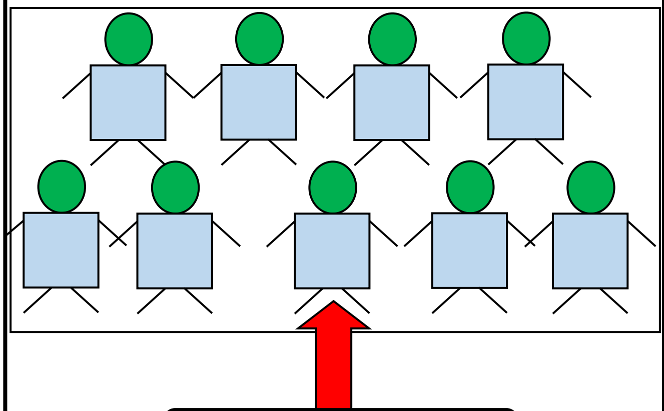
- Quantity supply can be represented by a point on the supply curve.



SUPPLY

Institutional

In a certain period of time the amount willing to supply at alternative prices by such as institution is known as institutional supply.



only firm

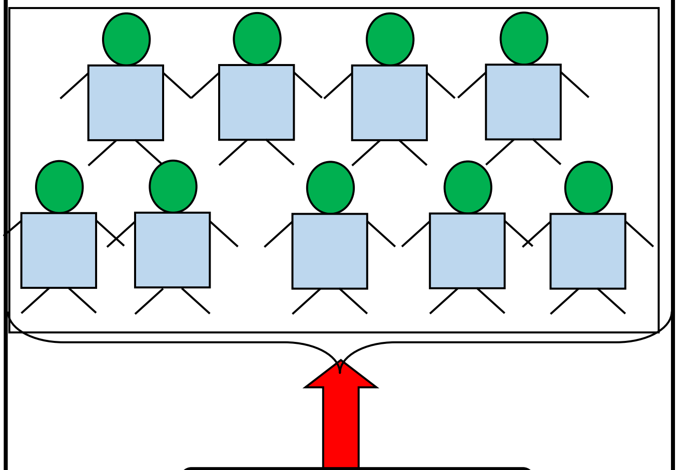
Determinants of Institutional supply.

- Price of the concerned goods(P)
- Price of the related goods(Pn)
- Consumer Income(Y)
- Consumers' Taste (T)
- Future expectations (Ex)
- Other factors (O)

Institutional supply function.

Market supply

The sum of total institutions product of a particular good or services is known as a market supply.



sum of every firms

Determinants of Market Supply

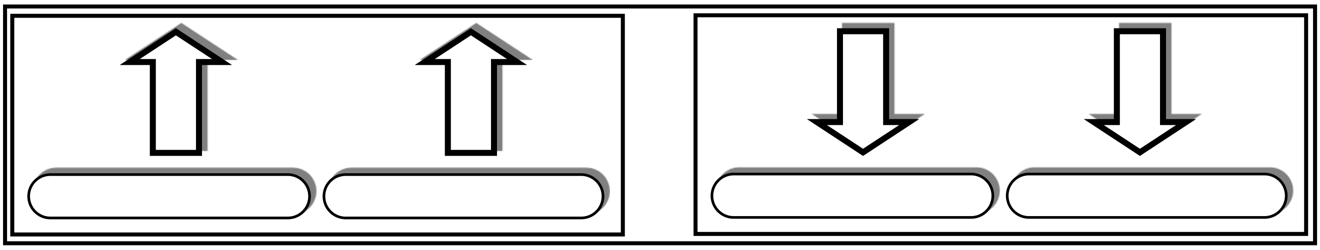
- Price of the concerned goods(P)
- Price of the related goods(Pn)
- Consumer Income(Y)
- Consumers' Taste (T)
- Number of firms (N)
- Future expectations (Ex)
- Other factors (O)

Market supply function.

Analysis the way of change in supply of a concerned good according to the change in determinants of supply is knows as theory of supply.

Law of Supply

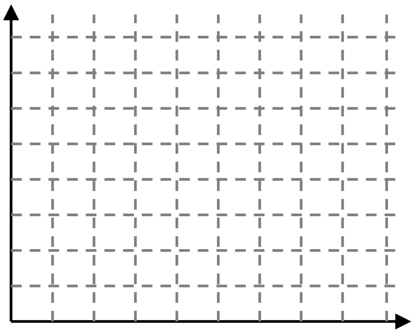
In a certain period of time, when other factors affecting supply remains constant the positive relationship between price and the quantity demand of goods, is called the law of supply.



The alternative methods which represent The Law of Demand are as follows.

- Supply schedule
- Supply curve
- Supply equation

Price	Quantity D
0	100
10	200
20	300
30	400

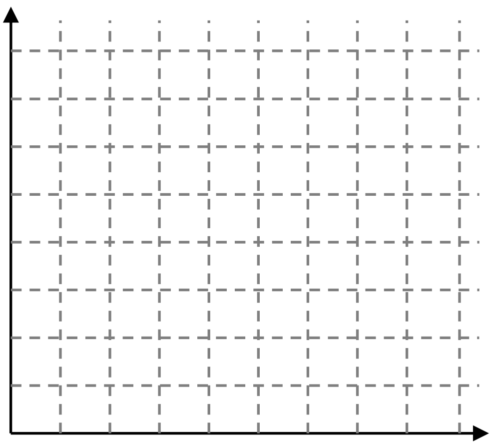


$$QS= a+bP$$

❍ Make the supply curve and estimate the supply equation.

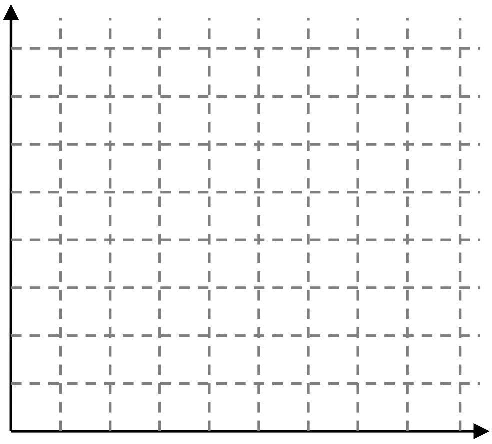
1)

මිල(P)	ප්‍රමාණය(Qs)
0	0
10	100
20	200
30	300



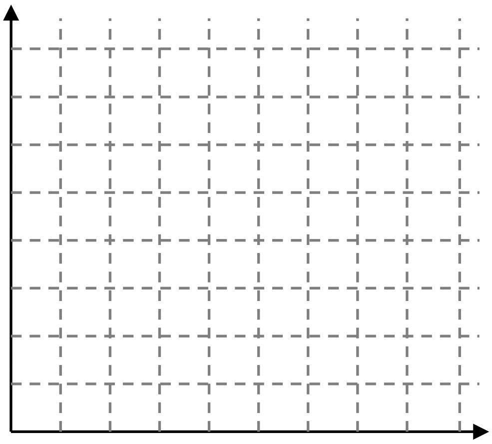
2)

මිල(P)	ප්‍රමාණය(Qs)
10	0
20	50
30	100
40	150



3)

මිල(P)	ප්‍රමාණය(Qs)
20	0
30	40
40	80
50	120



⏩ Reasons for the positive relationship between price and quantity Supply ⏩

❒ The concept of profit

The increase in the price of a product also increases the profit to the producer. Therefore, the manufacturer is directed to supply more in the hope of making more profit. On the other hand, when prices fall, the producer's profits fall. So when the price goes down, so does the supply.

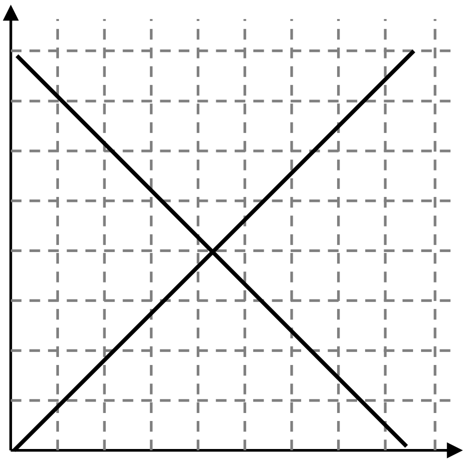
❒ The Law of Increasing Opportunity Cost

The Law of Increasing Opportunity Cost states that when the production of a particular good increases the opportunity cost also increases.

- Producers tend to produce more only if price increase easier to cover the cost because if they increase the production opportunity cost also increase.)

❒ Market equilibrium ❒

Market equilibrium is the situation where the expectations of purchasers and suppliers at a competitive market equal each other.



⏩ ⏩ The conditions of market equilibrium

- **Expected demand price should be equal to expected supply price.**
- **Expected quantity demanded should be equal to expected quantity supplied.**
- **Excess demand and excess supply must be zero.**
- **Expected demand price and expected supply price should be zero.**
- **Consumer expenditure and producer revenue should be equal to each other.**

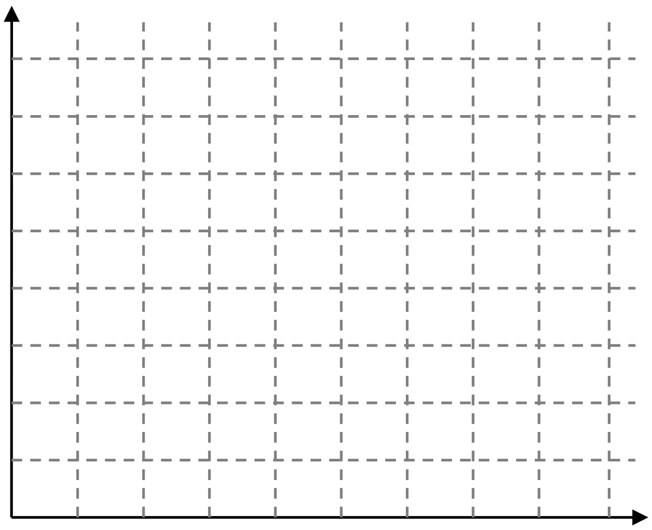
❒ The alternative methods which can be shown the market equilibrium ❒

1. By schedules of demand and supply

P	Qd	Qs
0	12	0
1	10	2
2	8	4
3	6	6
4	4	8
5	2	10
6	0	12

Demand Equation

2. By curves of demand and supply



Supply Equation

3. By equations of demand and supply

Find Equilibrium Price

Find Equilibrium Quantity

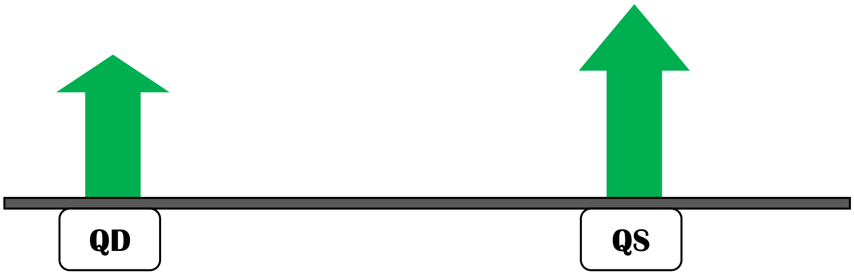
❒ Excess demand/ Excess supply

❒ Exess Demand ❒

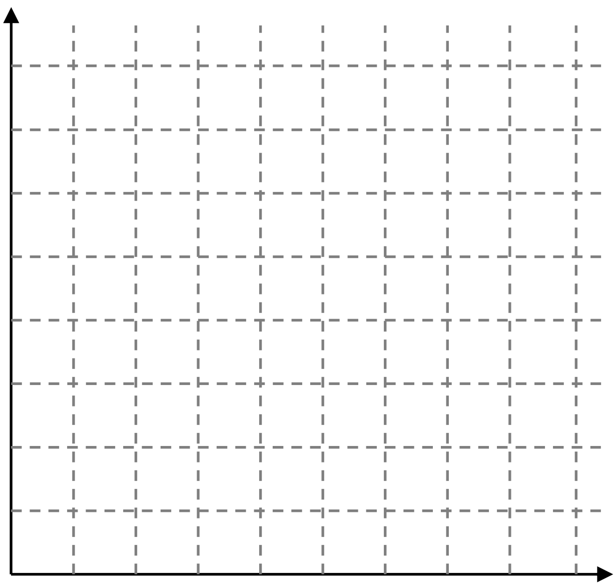
The amount of quantity demanded which exceeds the amount of quantity supplied at a certain price is called excess demand.

❒ Exess Supply ❒

The amount of quantity supplied which exceeds the amount of quantity demanded



P	Qd	Qs	ED_Q	ES_Q
0	12	0		
1	10	2		
2	8	4		
3	6	6		
4	4	8		
5	2	10		
6	0	12		



Excess demand and excess supply can also be explained with equations.

$Qd = 12 - 2p$ ← Demand equation

$Qs = 2p$ ← Supply equation

Excess demand Equation

Excess demand equation = demand equation – supply equation

According to the excess demand equation
excess demand can be calculated.

Example – In price Rs.3/-

Excess quantity demanded = 12 – 4p

= 12 - 4×3

= 12 -12 = 0

Excess Supply Equation

Excess supply equation = supply equation – demand equation

According to the excess supply equation excess supply can be calculated.

Exercise - 01

Qd= 4250-25P

QS= 2000+ 50P

01- Find equilibrium price.

QD =QS

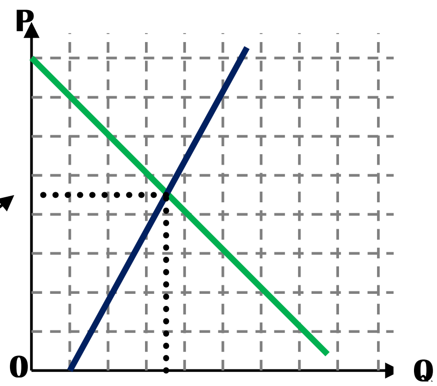
4250-25P = 2000+ 50P

4250-2000=50P+25P

2250 = 75P

$$\frac{2250}{75} = \frac{75P}{75}$$

30 = P



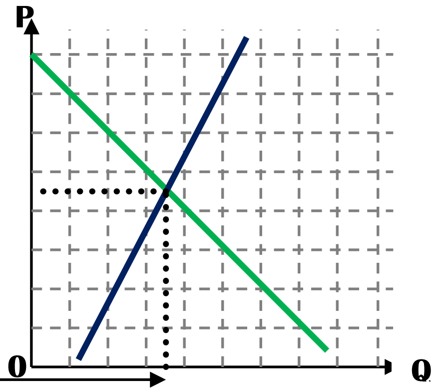
02- Find equilibrium quantity.

$$Qd = 4250 - 25\boxed{P}$$

$$Qd = 4250 - 25 \times 30$$

$$Qd = 4250 - 750$$

$$Qd = 3500$$



03- Find maximum demand price.

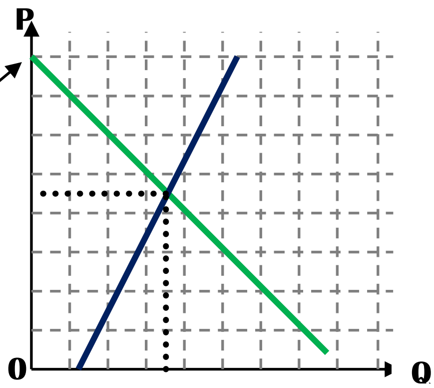
$$\boxed{Qd} = 4250 - 25P$$

$$0 = 4250 - 25P$$

$$25P = 4250$$

$$\frac{25P}{25} = \frac{4250}{25}$$

$$P = 170$$



Exercise - 02

Qd= 30-2P

QS= -6+ 2P

01- Find equilibrium price.

$$QD = QS$$

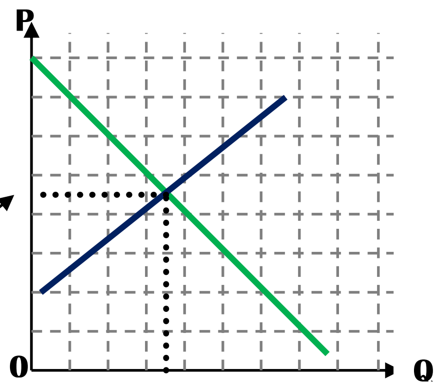
$$30-2P = -6 + 2P$$

$$30+6 = 2P + 2P$$

$$36 = 4P$$

$$\frac{36}{4} = \frac{4P}{4}$$

$$9 = P$$



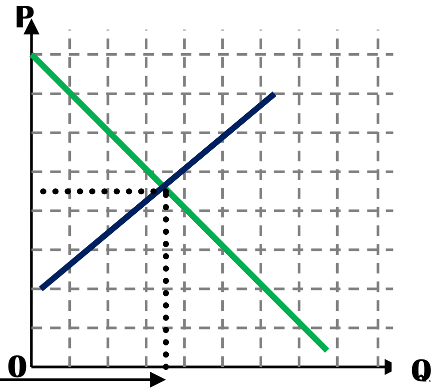
02- Find equilibrium quantity.

$$Qd = 30 - 2\boxed{P}$$

$$Qd = 30 - 2 \times 9$$

$$Qd = 30 - 18$$

$$Qd = 12$$



03- Find maximum demand price.

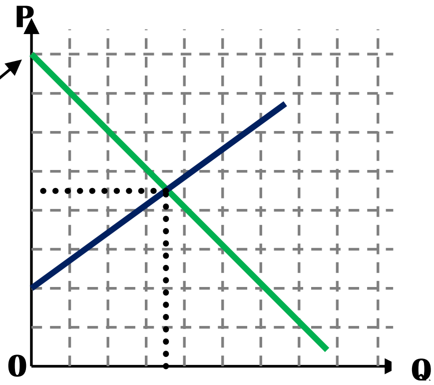
$$\boxed{Qd} = 30 - 2P$$

$$0 = 30 - 2P$$

$$2P = 30$$

$$\frac{2P}{2} = \frac{30}{2}$$

$$P = 15$$



04- Find minimum supply price.

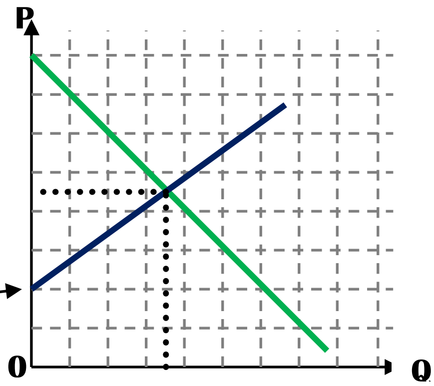
$$\boxed{Qs} = -6 + 2P$$

$$0 = -6 + 2P$$

$$6 = 2P$$

$$\frac{6}{2} = \frac{2p}{2}$$

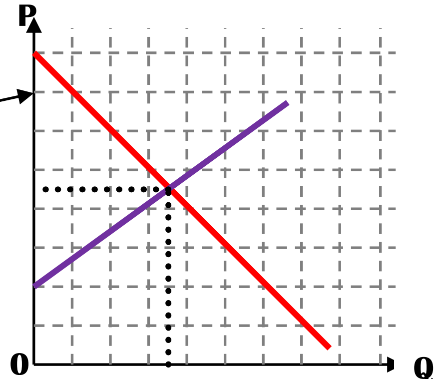
$$3 = P$$



Consumer surplus

Consumer Surplus is the difference between the price that consumers pay and the price that they are willing to pay.

Market equilibrium price is the price which consumer pays for goods and it not the price consumer is willing to pay.

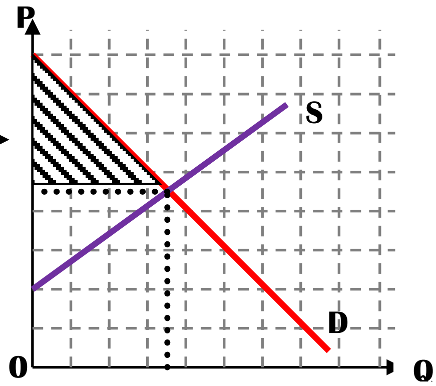


The price that consumer is willing to pay may be higher than that price

<u>Consumer surplus can be explained by a graph.</u>

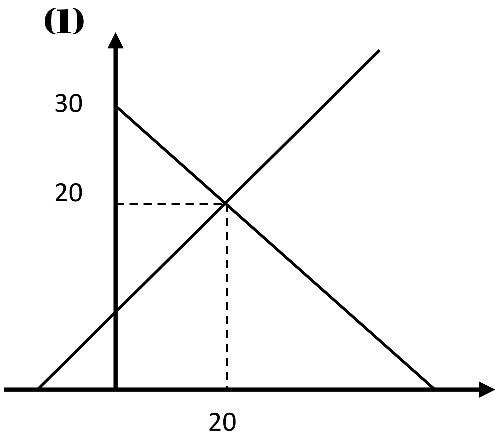
On a supply and demand curve, it is the area between the equilibrium price and the demand curve.

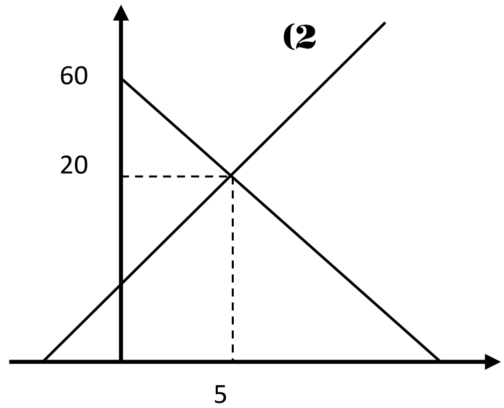
Consumer surplus



Consumer surplus can be calculated as follows

$$\text{Consumer surplus} = \frac{\left(\text{Maximum demand price} - \text{Equilibrium price}\right) \times \text{Equilibrium quantity}}{2}$$

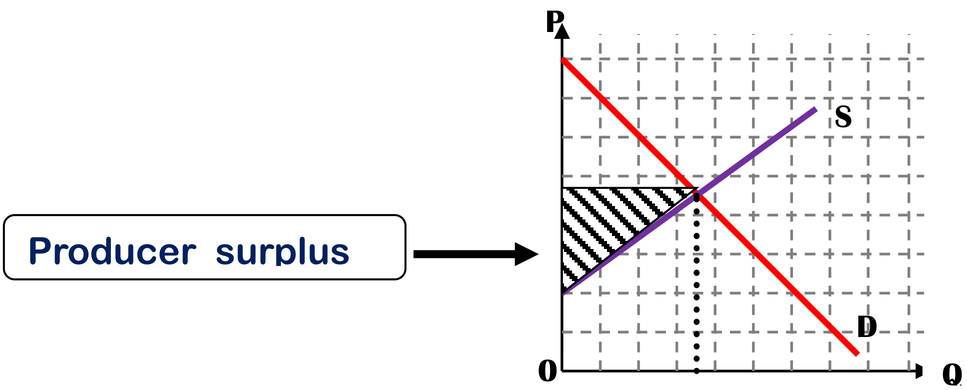




Producer surplus

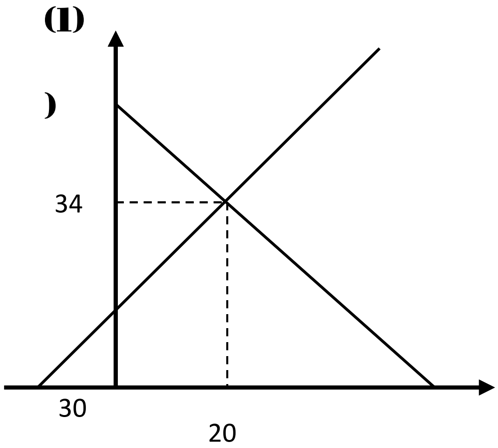
The difference between the minimum price that the supplier are willing to receive and the price they actually receive is known as producer surplus..

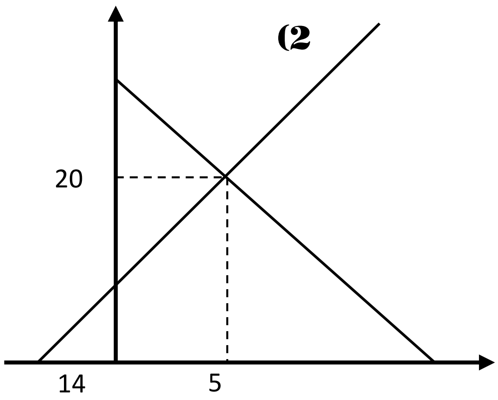
Producer surplus can be explained by a graph.



Producer surplus can be calculated as follows

$$\text{Producer surplus} = \frac{\left(\text{Equilibrium price - Minimum Supply price}\right) \times \text{Equilibrium quantity}}{2}$$





Economic surplus

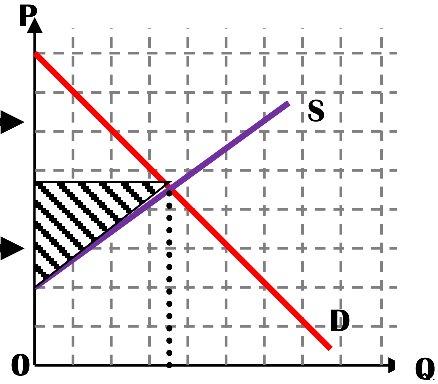
The buyers and sellers gained profits through the equilibrium exchange. The gain achieved by both parties is called economic surplus.

Economic surplus can be calculated as follows

$$\boxed{\text{Economic surplus} = \text{Consumer surplus} + \text{Producer surplus}}$$

<u>Economic surplus can be explained by a graph.</u>

Economic surplus
- Consumer surplus
- Producer surplus



Elasticity of demand

Measure the responsiveness of the quantity demanded to the changes any of the determinants of demand, which can be measured quantitatively, is named as elasticity of demand.

$$Qd = f\ (P\ ,\quad P_n,\quad Y)$$

P	P_n	Y
Price elasticity of demand	Cross Price elasticity of demand	Income elasticity of demand

Price elasticity of demand

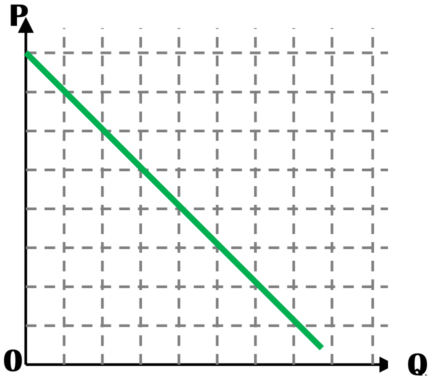
Measure the percentage change of quantity demanded in response to the percentage change of price is identified as price elasticity of demand.

$$Qd = f\ (P)$$

There are two ways of calculating the price elasticity of demand.

- **Point price elasticity of demand**

When other determinants of demand remain constant measure the percentage change in quantity demanded according to the small percentage change in price at a particular point on the demand curve is known as point price elasticity of demand.



The following formula used to measure the point price elasticity

Calculate a percentage change in price that affects the Quantity of demand.

$$\text{Point Price elasticity of demand} = \frac{\text{Percentage change in quantity demand}}{\text{Percentage change in price}} = \frac{\Delta Qd\ \%}{\Delta P\ \%}$$

EXERCISE - 01

01- When the price of a commodity increases by 10% quantity demand decreases by 20%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\Delta Qd\ \%}{\Delta P\ \%}$$

02- When the price of a commodity increases by 20% quantity demand decreases by 40%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\blacktriangle Qd\,\%}{\blacktriangle P\,\%}$$

03- When the price of a commodity increases by 15% quantity demand decreases by 30%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\blacktriangle Qd\,\%}{\blacktriangle P\,\%}$$

04- When the price of a commodity increases by 40% quantity demand decreases by 20%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\blacktriangle Qd\,\%}{\blacktriangle P\,\%}$$

05- When the price of a commodity increases by 2% quantity demand decreases by 1%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\blacktriangle Qd\,\%}{\blacktriangle P\,\%}$$

06- When the price of a commodity increases by 3% quantity demand decreases by 6%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\blacktriangle Qd\,\%}{\blacktriangle P\,\%}$$

07- When the price of a commodity increases by 4% quantity demand decreases by 100%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\blacktriangle Qd\,\%}{\blacktriangle P\,\%}$$

08- When the price of a commodity increases by 3% quantity demand decreases by 15%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\Delta Qd\,\%}{\Delta P\,\%}$$

09- When the price of a commodity increases by 10% quantity demand decreases by 4%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\Delta Qd\,\%}{\Delta P\,\%}$$

10- When the price of a commodity increases by 5% quantity demand decreases by 25%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\Delta Qd\,\%}{\Delta P\,\%}$$

11- When the price of a commodity increases by 20% quantity demand decreases by 60%. Calculate elasticity of demand.

$$\text{Point Price elasticity of demand} = \frac{\Delta Qd\,\%}{\Delta P\,\%}$$

- **The following formula used to measure the point price elasticity**

Calculating elasticity of demand when given a price schedule.

$$\text{Point Price elasticity of demand} = \frac{\Delta Qd}{\Delta P} \times \frac{P}{Qd}$$

EXERCISE 02

1)

P	QD
10	60
20	40

Calculate the elasticity of demand at the price of Rs. 10

Calculate the elasticity of demand at the price of Rs. 20

2)

P	QD
4	400
6	200

Calculate the elasticity of demand at the price of Rs. 4.

Calculate the elasticity of demand at the price of Rs. 6.

3)

P	QD
5	100
10	75

Calculate the elasticity of demand at the price of Rs. 5.

Calculate the elasticity of demand at the price of Rs. 10.

4)

P	QD
10	100
20	50

Calculate the elasticity of demand at the price of Rs. 10.

Calculate the elasticity of demand at the price of Rs. 20.

5)

P	QD
4	160
8	120

Calculate the elasticity of demand at the price of Rs. 4.

Calculate the elasticity of demand at the price of Rs. 8

6)

P	QD
20	400
40	200

Calculate the elasticity of demand at the price of Rs. 20.

Calculate the elasticity of demand at the price of Rs. 40.

7)

P	QD
4	200
6	100

Calculate the elasticity of demand at the price of Rs. 4.

Calculate the elasticity of demand at the price of Rs. 6.

8)

P	QD
12	80
16	40

Calculate the elasticity of demand at the price of Rs. 12.

Calculate the elasticity of demand at the price of Rs. 16.

9)

P	QD
120	800
140	700

Calculate the elasticity of demand at the price of Rs. 120.

Calculate the elasticity of demand at the price of Rs. 140.

10)

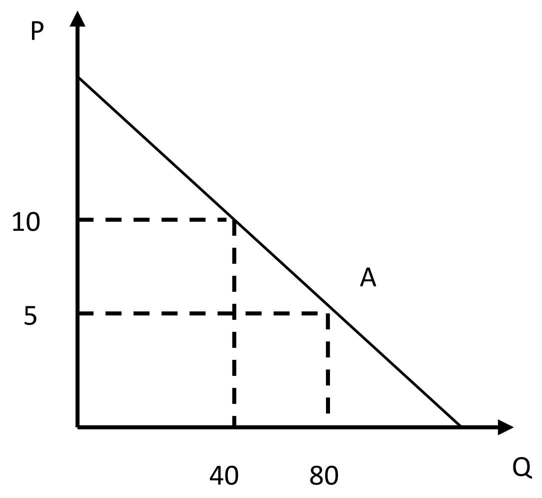
P	QD
240	720
280	600

Calculate the elasticity of demand at the price of Rs. 240.

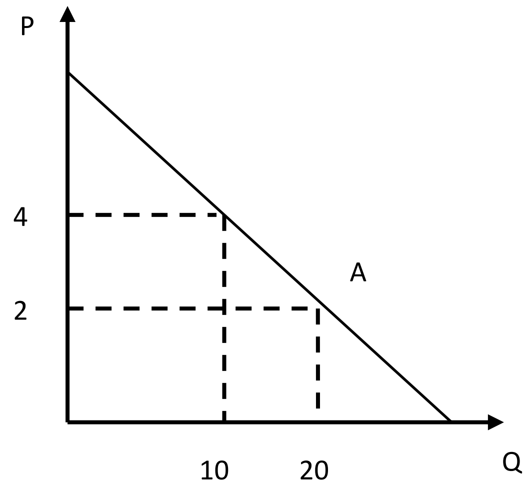
Calculate the elasticity of demand at the price of Rs. 280.

EXERCISE 03

1) Calculate the price elasticity of demand at the relevant point in the graph below.



2)



The following formula used to measure the point price elasticity

Calculating elasticity of demand when given a Demand equation.

$$\text{Point Price elasticity of demand} = \boxed{\frac{\Delta Qd}{\Delta P}} \times \frac{P}{Qd}$$

b

$$\text{Point Price elasticity of demand} = \boxed{-b} \times \frac{P}{Qd}$$

EXERCISE 04

Qd=500-5P Calculate the price elasticity of demand at the price Rs. 20.

STEP 01	**STEP 02**

1) Qd=20-3P Calculate the price elasticity of demand at the price Rs. 2.

2) Qd=100-4P Calculate the price elasticity of demand at the price Rs. 10.

3) Qd=100-4P Calculate the price elasticity of demand at the price Rs. 20.

4) Qd=200-5P Calculate the price elasticity of demand at the price Rs. 10.

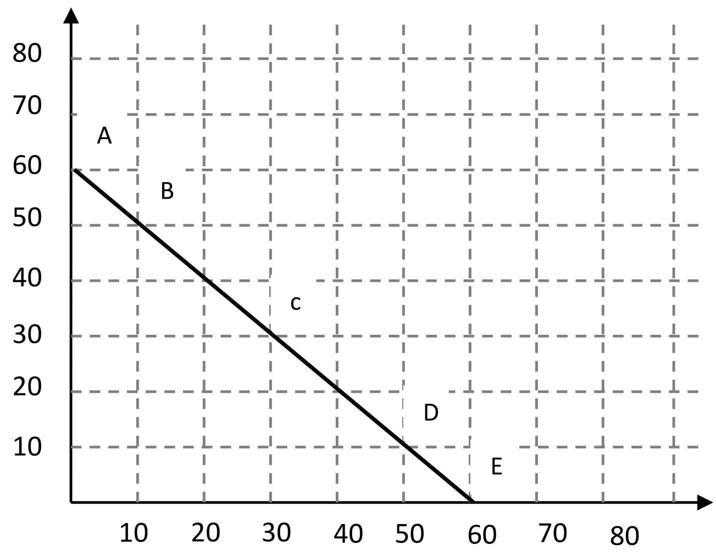
5) Qd=50-10P Calculate the price elasticity of demand at the price Rs. 2

6) Qd=50-10P Calculate the price elasticity of demand at the price Rs. 3.

7) $Qd=50-5P$ Calculate the price elasticity of demand at the price Rs. 6

✪ Calculating the elasticity of different points on the demand curve

P	QD
0	60
10	50
20	40
30	30
40	20
50	10
60	0



$$P.E.D = \frac{\Delta Qd}{\Delta P} \times \frac{Qd}{P}$$

Elasticity	Value of coefficient of elasticity	Change in price	Consumer Expenditure/ Producer revenue
Perfect elastic	PED= ∞	Increase	Zero
		Decrease	Infinity
Elastic	PED > 1	Increase	Decrease
		Decrease	Increase
Unitary	PED = 1	Increase	Constant
		Decrease	Constant
Inelastic	PED < 1	Increase	Increase
		Decrease	Decrease
Perfectly Inelastic	PED = 0	Increase	Increase
		Decrease	Increase

Determinants of price elasticity of demand

1. Nature of the concerned good as essential or as luxury of the concerned good

2. Number of substitutes available to a good and its closeness.

3. Percentage of the income spent on the good.

4. The alternative uses of goods

5. The time which take to adjust to the price change

The practical importance of price elasticity of demand

- The concept of elasticity is important to make decisions for economic policy compilers, producers and consumers.
- It can be predict the effect for consumer expenditure or producer revenue when price increase or decrease.
- Determination of when out put income is maximized.
- Monopoly power can be recognized in a business firm
- Can be used to compile the economic policies.
- To choose concerned goods when imposing taxes.

Cross price elasticity demand

When other factors affecting demand constant which measures the responsiveness of the change of quantity demand of the concerned goods to the change of the price of relative goods, is known as cross price elasticity of demand.

1. **When the price of a good increases by 10%, the quantity demand of the concerned good increases by 40%. Calculate cross elasticity of demand.**

2. **When the price of a good increases by 10%, the quantity demand of the concerned good decreases by 30%. Calculate cross elasticity of demand.**

3. **When the price of a good increases by 10%, the quantity demand of the concerned good increases by 30%. Calculate cross elasticity of demand.**

4. **When the price of a good decreases by 20%, the quantity demand of the concerned good increases by 40%. Calculate cross elasticity of demand.**

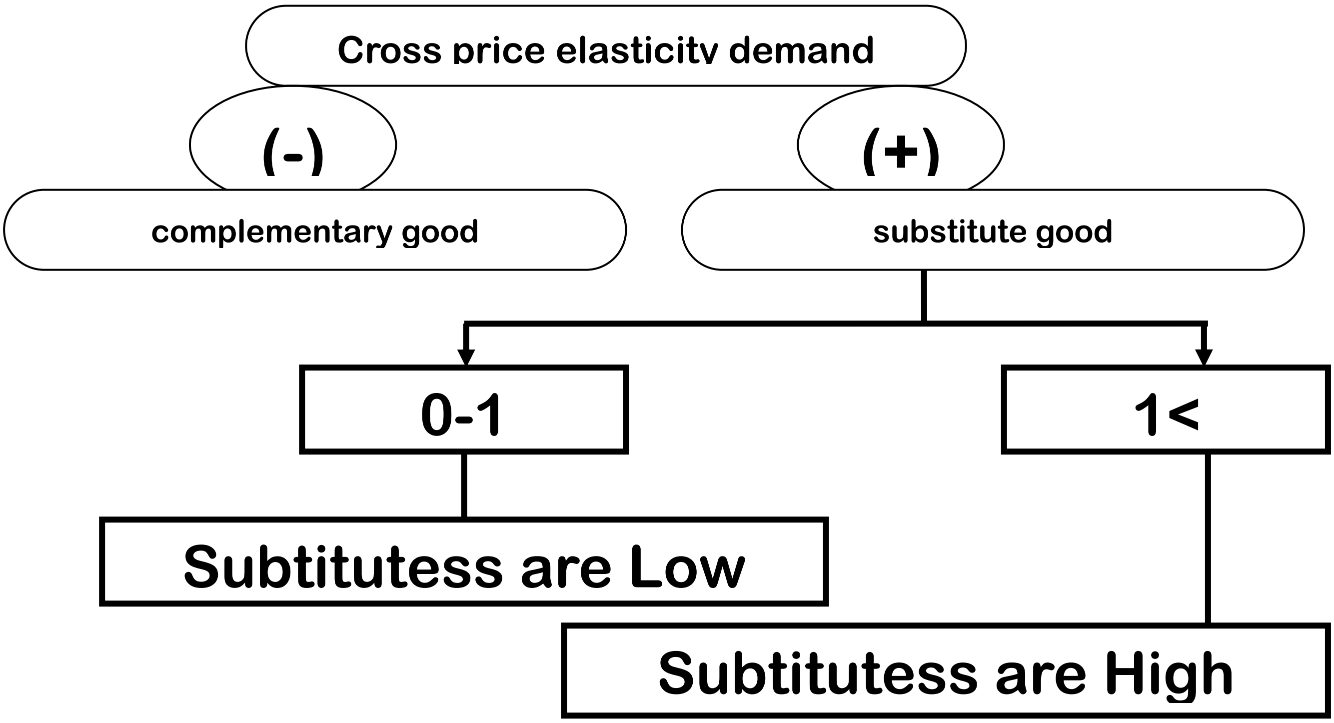
5. **When the price of a good decreases by 10%, the quantity demand of the concerned good increases by 80%. Calculate cross elasticity of demand.**

6. **When the price of a good decreases by 20%, the quantity demand of the concerned good increases by 60%. Calculate cross elasticity of demand.**

Several types of goods can be identified based on cross-demand flexibility values

If there is a direct relationship between the price of related goods and the demand of the concerned goods the co-efficient will get a positive value and the good is considered as a substitute for the concerned goods.

If there is a negative (inverse) relationship exits between the price of the related goods and the demand of the concerned goods the coefficient will get a negative value and the good is considered as a complementary good to the concerned good.



The practical important of cross price elasticity

- Analyse the inter-relationship between two goods.

- Determine the market competition of goods

- Forecast the relative changes in demand for goods and services

Income elasticity demand

When other factors affecting demand remain constant measure the responsiveness of the percentage change of demand to a percentage change of income is known as income elasticity of demand.

$$IED = \frac{\text{Change in demand}}{\text{Change in income}} \qquad \frac{\Delta QD}{\Delta Pn}$$

1. **When a person's income increases by 20%, the quantity demand of the concerned good decreases by 40%. Calculate the Income elasticity of demand.**

2. **When a person's income increases by 30%, the quantity demand of the concerned good decreases by 60%. Calculate the Income elasticity of demand.**

3. **When a person's income increases by 20%, the quantity demand of the concerned good decreases by 80%. Calculate the Income elasticity of demand.**

Several types of goods can be identified based on cross-demand flexibility values

Cross price elasticity demand

(-) — **inferior good**

(+) — **normal goods**

0-1	1<
essential good	luxury good

The practical importance of income elasticity of demand

- *goods can be classified as essential, luxury and inferior goods*
- *can be forecast, how the demand changes according to the changes in income.*

Elasticity Of supply

The responsiveness of quantity supplied to the change in the price of concerned good, when other determinants remain constant is the price elasticity of supply.

Price elasticity of Supply

Measure the percentage change of quantity supply in response to the percentage change of price is identified as price elasticity of supply.

$$Qs = f\ (P)$$

The following formula used to measure the price elasticity

Calculate a percentage change in price that affects the Quantity of supply.

EXERCISE - 01

01- When the price of a commodity increases by 10% quantity supply increases by 20%. Calculate elasticity of supply.

$$\text{Point Price elasticity of demand} = \frac{\Delta Qs\,\%}{\Delta P\,\%}$$

02- When the price of a commodity increases by 20% quantity supply increases by 40%. Calculate elasticity of supply.

Point Price elasticity of demand $= \dfrac{\blacktriangle Qs\,\%}{\blacktriangle P\,\%}$

03- When the price of a commodity increases by 15% quantity supply increases by 30%. Calculate elasticity of supply.

Point Price elasticity of demand $= \dfrac{\blacktriangle Qs\,\%}{\blacktriangle P\,\%}$

04- When the price of a commodity increases by 40% quantity supply increases by 20%. Calculate elasticity of supply.

Point Price elasticity of demand $= \dfrac{\blacktriangle Qs\,\%}{\blacktriangle P\,\%}$

05- When the price of a commodity increases by 2% quantity supply increases by 1%. Calculate elasticity of supply.

Point Price elasticity of demand $= \dfrac{\blacktriangle Qs\,\%}{\blacktriangle P\,\%}$

06- When the price of a commodity increases by 3% quantity supply increases by 6%. Calculate elasticity of supply.

Point Price elasticity of demand $= \dfrac{\blacktriangle Qs\,\%}{\blacktriangle P\,\%}$

07- When the price of a commodity increases by 4% quantity supply increases by 100%. Calculate elasticity of supply.

Point Price elasticity of demand $= \dfrac{\blacktriangle Qs\,\%}{\blacktriangle P\,\%}$

08- When the price of a commodity increases by 3% quantity demand increases by 15%. Calculate elasticity of supply.

$$\text{Point Price elasticity of demand} = \frac{\blacktriangle Qs\ \%}{\blacktriangle P\ \%}$$

09- When the price of a commodity increases by 10% quantity demand increases by 4%. Calculate elasticity of supply.

$$\text{Point Price elasticity of demand} = \frac{\blacktriangle Qs\ \%}{\blacktriangle P\ \%}$$

The following formula used to measure the point price elasticity

Calculating elasticity of supply when given a price schedule.

$$\text{Point Price elasticity of supply} = \frac{\blacktriangle Qs}{\blacktriangle P} \times \frac{P}{Qs}$$

EXERCISE 02

1)

P	Qs
10	40
20	60

Calculate the elasticity of supply at the price of Rs. 10

Calculate the elasticity of supply at the price of Rs. 20

2)

P	Qs
4	200
6	400

Calculate the elasticity of supply at the price of Rs. 4.

Calculate the elasticity of supply at the price of Rs. 6.

3)

P	Qs
5	75
10	100

Calculate the elasticity of supply at the price of Rs. 5.

Calculate the elasticity of supply at the price of Rs. 10.

4)

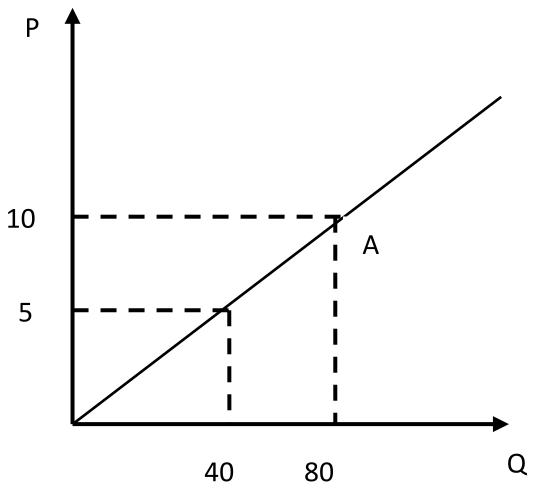
P	Qs
10	50
20	100

Calculate the elasticity of supply at the price of Rs. 10.

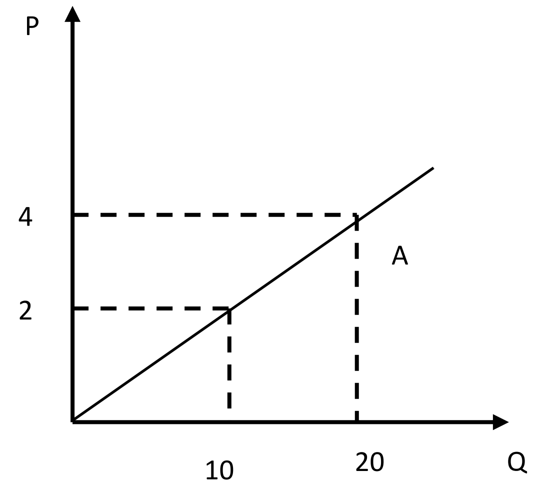
Calculate the elasticity of supply at the price of Rs. 20.

EXERCISE 03

1) Calculate the price elasticity of demand at the relevant point in the graph below.



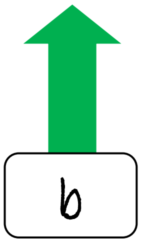
2)



- **The following formula used to measure the point price elasticity**

Calculating elasticity of supply when given a supply equation.

$$\text{Point Price elasticity of supply} = \boxed{\frac{\Delta Q_s}{\Delta P}} \times \frac{P}{Q_s}$$



$$\text{Point Price elasticity of supply} = \boxed{b} \times \frac{P}{Qs}$$

EXERCISE 04

Qs=500+5P Calculate the price elasticity of supply at the price Rs. 20.

STEP 01 **STEP 02**

1) Qs=20+3P Calculate the price elasticity of supply at the price Rs. 2.

2) Qs=100+4P Calculate the price elasticity of supply at the price Rs. 10.

3) Qs=100+4P Calculate the price elasticity of demand at the price Rs. 20.

4) Qs=200+5P Calculate the price elasticity of demand at the price Rs. 10.

5) Qs=50+10P Calculate the price elasticity of demand at the price Rs. 2

6) Qs=50+10P Calculate the price elasticity of demand at the price Rs. 3.

7) Qs=50+5P Calculate the price elasticity of demand at the price Rs. 6

8) Qs=50+5P Calculate the price elasticity of demand at the price Rs. 5

9) Qs=200+5P Calculate the price elasticity of demand at the price Rs. 12.

10) Qs=20+2P Calculate the price elasticity of demand at the price Rs. 5.

✪ Elasticity of supply